UI/UX & Technical Design Document

Project: Automated Skeletal Analysis System

Platform: Web Application (Desktop-optimized dashboard)

1. System Overview

The Automated Skeletal Analysis System is a specialized dashboard designed for forensic anthropologists, medical examiners, or researchers. It allows users to input skeletal measurements, run predictive models to determine demographic profiles (age, gender, height), and manage past forensic cases.

2. Technology Stack

- **Frontend:** React.js, Vite (Bundler), Tailwind CSS (Styling)
- **Backend:** Node.js, Express.js (RESTful API handling business logic and model integration)
- **Database:** Supabase (PostgreSQL for structured case data, user auth, and potentially storage for image guides/reports)
- **Suggested Libraries:** * react-router-dom (Routing)
  - recharts or chart.js (Data visualization)
  - lucide-react or heroicons (UI Icons)
  - react-hook-form (Form handling)

---

3. Design System & UI Guidelines

3.1 Color Palette (Dark Theme Default)

The application utilizes a dark, modern interface suitable for prolonged analytical work, reducing eye strain.

- **Background (Main):** #1E222D (Dark Slate Blue)
- **Background (Sidebar/Cards):** #252A36 (Slightly lighter slate)
- **Primary Accent:** #3B82F6 (Vivid Blue - used for active states, primary buttons, charts)
- **Secondary Accents:**
  - Success/Step Active: #10B981 (Emerald Green)
  - Warning/Back/Delete: #EF4444 (Red)
- **Text Colors:**
  - Primary Text: #F3F4F6 (Off-White)
  - Secondary Text: #9CA3AF (Cool Gray)

3.2 Typography

- **Font Family:** Inter or Roboto (Standard modern sans-serif).
- **Hierarchy:**
    - H1 (Dashboard Titles): 24px, Semi-Bold, White
    - H2 (Section Headers): 18px, Medium, White
    - Body Text: 14px, Regular, Light Gray
    - Small/Labels: 12px, Regular, Cool Gray

3.3 Reusable React Components

To maintain Vite+React best practices, the following UI elements should be built as reusable components:

- <Sidebar />: Navigation wrapper with active state styling.
- <StatCard />: Displays icon, title, value, and subtitle (Used in Dashboard).
- <DataTable />: Reusable table component with props for columns, data, search, and pagination.
- <Stepper />: Visual progress indicator for the multi-step form.
- <FormInput /> / <FormSelect />: Standardized input fields with Tailwind styling.
- <Button />: Configurable variants (Primary/Blue, Secondary/Red, Success/Green, Outline).

---

4. Screen-by-Screen Breakdown & UX Flow

Screen 1: Main Dashboard (Dashboard1.JPG)

- **Purpose:** Provide a high-level overview of system metrics and recent activity.
- **Layout:**
    - **Top Row:** Four <StatCard /> components (Total Cases, Recent Analysis, Avg Accuracy, Last Prediction).
    - **Middle Row:** * Age Distribution Bar Chart (Needs a charting library).
        - Gender Distribution Donut Chart.
    - **Bottom Row:** Recent Cases <DataTable /> with a search bar and sort dropdown. Includes View (Eye) and Delete (Trash) action icons.
- **UX Note:** Hover tooltips should be implemented on charts to show exact numbers.

Screen 2: New Analysis - Step 1 (Dashboard2.JPG)

- **Purpose:** Initiate a new case by collecting basic metadata.
- **Layout:**
    - **Top:** <Stepper currentStep={1} /> (1. Basic Information, 2. Skeletal Measurements, 3. Review & Predict).
    - **Body:** A 2-column grid form for basic info (Case ID, User Name, Bones Type, Location, Dates, Email).
    - **Action:** "Next" button aligned to the bottom right.

- **Tech Note:** Use react-hook-form to capture and validate this data. Store the state in a React Context or Redux store so it persists when moving to Step 2.

Screen 3: New Analysis - Step 2 (Dashboard3.JPG)

- **Purpose:** Input specific skeletal parameters required for the prediction model.
- **Layout:**
  - **Top:** <Stepper currentStep={2} />
  - **Body (Upper):** "Skull Measurements" form. Uses standardized <FormSelect /> dropdowns (Brow Ridge, Jaw Shape, Nuchal Crest, Mastoid Size, Cranial Suture).
  - **Body (Lower):** "Guide For Measurements" panel. Contains explanatory text and reference images to ensure user accuracy.
  - **Actions:** "Back" (Red) and "Next" (Blue) buttons.
- **UX Note:** The guide images are crucial for scientific accuracy. Implementing a feature where clicking the image opens a larger lightbox view would enhance UX.

Screen 4: New Analysis - Step 3 (Dashboard4.JPG)

- **Purpose:** Display the results of the automated prediction.
- **Layout:**
  - **Top:** <Stepper currentStep={3} />
  - **Middle:** Four highlight cards showing the prediction: Gender, Age range, Height, and Confidence Level.
  - **Lower:** "Similar Cases" <DataTable />. This cross-references the database for historical cases with similar skeletal traits.
  - **Actions:** "Back To Dashboard" (Green) and "Generate Report" (Blue).
- **Tech Note:** Reaching this screen triggers a POST request to the Node.js/Express backend, which processes the inputs, communicates with the ML model, stores the new case in Supabase, and returns the result to the React frontend.

Screen 5: Prediction Report (Dashboard5.JPG)

- **Purpose:** A comprehensive summary view for export or final review.
- **Layout:**
  - **Top Actions:** "Send Mail" (Green) and "Download Report" (Blue).
  - **Grid Layout:** * Left Column: Basic Information and Skeletal Features summary.
    - Right Column: Prediction Result and Confidence Level.
  - **Lower:** Similar Cases table and Age Distribution chart specifically related to similar cases.
- **UX Note:** The "Download Report" function should trigger a PDF generation utility (like jspdf or html2pdf.js on the frontend, or a backend PDF generator).

Screen 6: Past Analysis (Dashboard6.JPG)

- **Purpose:** A searchable repository of all historical case data.

- **Layout:**
  - Full-width <DataTable /> component.
  - Includes Search input at the top right.
  - Pagination controls at the bottom right.
- **Tech Note:** The React component should fetch data from the Express backend, querying the Supabase database. Server-side pagination and filtering should be implemented to handle large datasets (e.g., 256K entries) efficiently.

---

5. Frontend Architecture (React + Vite + Tailwind)

Suggested Folder Structure

Plaintext

```
src/
├── assets/          # Images (Guide images, logos)
├── components/      # Reusable UI components
│   ├── layout/      # Sidebar, Header, LayoutWrapper
│   ├── common/      # Button, Input, Select, Stepper
│   ├── data/        # DataTable, StatCard
│   └── charts/      # BarChart, DonutChart wrappers
├── pages/           # Route components
│   ├── Dashboard.jsx
│   ├── NewAnalysis/
│   │   ├── Step1BasicInfo.jsx
│   │   ├── Step2Measurements.jsx
│   │   └── Step3Review.jsx
│   ├── Report.jsx
│   └── PastAnalysis.jsx
├── context/         # AnalysisFormContext (State management across steps)
├── services/        # API calls (axios instances calling Node.js backend)
├── utils/           # Helper functions (date formatting, etc.)
├── App.jsx          # Router setup
└── index.css        # Tailwind directives & global styles
```

Tailwind Configuration Outline (tailwind.config.js)

To easily manage the dark theme, extend the Tailwind theme:

JavaScript

```javascript
module.exports = {
  content: ["./index.html", "./src/**/*.{js,ts,jsx,tsx}"],
  theme: {
    extend: {
      colors: {
        dark: {
          main: '#1E222D',   // Main background
          card: '#252A36',   // Cards and Sidebar
          border: '#374151', // Subtle borders
        },
        brand: {
          blue: '#3B82F6',
          green: '#10B981',
          red: '#EF4444'
        }
      }
    },
  },
  plugins: [],
}
```

6. Backend Integration Guidelines (Node/Express + Supabase)

- **Supabase Schema:** You will need tables for cases (Basic Info + Predictions), measurements (Foreign key to case ID), and users (handled via Supabase Auth).
- **Express Routes Needed:**
  - GET /api/dashboard/stats: Fetch aggregate data for the main dashboard.
  - POST /api/analysis/predict: Accepts Step 1 & 2 data, processes prediction, saves to Supabase, returns Step 3 results.
  - GET /api/cases: Fetch past analyses with query params for pagination and search (?page=1&limit=40&search=keyword).
  - GET /api/cases/:id/similar: Fetch similar cases based on specific skeletal parameters.